greendash

The green recommendation that shows its receipts · M2: Progress · Green Customer Insights Platform (Lloyds)

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1. Current approach and changes since M1

The core architecture is unchanged from M1: pure-function engines behind typed Pydantic contracts, a deterministic carbon engine, rules-based eligibility, and an LLM that only explains. The table below records what was shipped or hardened since M1 was submitted.

Component	Status	What changed / was completed
Categoriser (FR2)	Shipped	Ordered keyword rules + TF-IDF character n-gram LR fall-through. Combined Macro-F1 > 0.91 on 200-string hand-checked gold set.
Propensity models (FR3)	Shipped	4 LightGBM models (GP01–GP04), isotonic calibration, SHAP grouped into 5 fixed reason codes (HIGH_FUEL_SPEND, LOW_ELEC_KWH, MISSING_DATA, NO_CAR, HOMEOWNER).
Confidence modelling	Shipped	Isolation Forest OOD detector + bootstrapped stability. Brier score falls T1→T3 (0.045 / 0.080 / 0.120), confirming the confidence signal is monotone.
LLM gateway + Claim-Check (FR5)	Shipped	Gemini at temp 0, allow-list filtered. Verifier blocks new numbers, certainty phrases, product names not in engine output; fails safely to a template.
Backend API (FR7)	Shipped	FastAPI, JWT + role-based access, CORS allowlist, object-level ownership. 5 endpoints tested: GET/POST /transactions, GET /transactions/{id}, POST /import/raw, POST /import/simple.
Privacy controls	Hardened	PII scrubber (regex + Luhn), HMAC-SHA256 pseudonymisation, LLM payload sanitiser. Verified no raw strings, names, or postcodes reach the model.
Data quality gates	All PASS	Schema (pandera), FK integrity, take-up base rates, single-feature AUC ceiling, 8/8 persona facts, SHA-256 reproducibility, PII scan.
Recommendation orchestrator	In progress	Status rules RULED_OUT > REVIEW > ABSTAIN > RECOMMEND; Evidence Pack builder.
Streamlit UI	In progress	Customer Hub (Proof Cards) and Cockpit drill-down layout started; routing pending.
Scenario Lab	Not started	Live data-mutation + recompute with displayed cause. Starts after orchestrator.

2. Experiments run so far and their results

2a. Data quality gates — all PASS (python -m Code.eval.data_quality, seed 42, n=5,000)

Gate	Detail	Threshold	Result
Schema conformance (pandera)	5 tables, 0 failures	100 %	PASS
Foreign-key integrity	7 tables, 0 orphans	0 orphans	PASS
Take-up base rates	GP01 5.1 % (el. 6.4 %), GP02 3.4 % (el. 4.8 %), GP03 2.8 % (el. 4.0 %), GP04 5.6 % (el. 18.4 %)	Inside target ranges	PASS
Single-feature AUC ceiling	GP01 0.700, GP02 0.716, GP03 0.691, GP04 0.560	All < 0.80	PASS
Persona signal facts	8 of 8 matched	8 of 8	PASS
Same-seed reproducibility	10 files; SHA-256 identical	Identical	PASS
PII scan on public data	0 hits	0 hits	PASS

2b. Categoriser — 200-string gold set | 2c. Confidence tercile validation

System	Macro-F1
Rules only (baseline)	~0.82
TF-IDF LR only	~0.79
Combined (rules → LR)	>0.91

Tercile	Brier	Criterion
T1 — Low	0.045	Monotone ✓
T2 — Med	0.080	Monotone ✓
T3 — High	0.120	Low→high ✓

2d. Propensity models — test set (15 % held out, fixed seed 42)

Product	ROC-AUC	PR-AUC	Brier	Lift@10 %	Status
GP01 — EV Finance	0.81	0.34	0.040	3.1×	Isotonic ✓
GP02 — Green Home	0.79	0.29	0.031	2.8×	Isotonic ✓
GP03 — Solar Finance	0.77	0.25	0.026	2.6×	Isotonic ✓
GP04 — Paperless Banking	0.71	0.41	0.052	2.2×	Isotonic ✓

LightGBM beats logistic-regression and rule-heuristic baselines on ROC-AUC and Lift@10 % for all four products. Monotone constraints enforced (higher fuel spend → higher EV propensity).

2e. LLM Claim-Check firewall — 100-pack evaluation | 2f. Carbon engine unit tests

Criterion	Obs.	Target	Result
Faithfulness (no new numbers)	100 %	≥95 %	PASS
Positive tone	100 %	≥90 %	PASS
Canary / PII leakage	0 %	0 %	PASS
Fallback on firewall failure	Yes	Req'd	PASS

Six hand-calculated unit tests covering all four product formulas (EV switch, Green Home efficiency, Solar install, Paperless). All 6 PASS; factor vintage pinned to source workbook row; INSUFFICIENT_DATA returned when inputs are absent.

3. Progress on testing

pytest Code/tests/unit/ -q — **60 passed, 1 skipped, 0 failures.**

Test file	P	S	Coverage area
test_api.py	15	0	Auth boundary, schema validation, ownership isolation, CSV import + atomic rollback
test_backend_db.py	17	0	Schema, FK enforcement, decimal precision, date validation, atomic rollback
test_config.py	14	0	Prod fail-closed, secret non-exposure, CORS allowlist, JWT key length
test_carbon.py	6	0	Hand-calculated formulas, factor pinning, INSUFFICIENT_DATA path
test_privacy.py	7	0	PII scrub, non-Luhn false positives, LLM payload sanitiser, pseudonymisation
test_templates.py	1	0	Claim-Check firewall on all LLM fallback templates
test_factors.py	0	1	Factor workbook check (workbook not in repo — skipped)

Still to test: orchestrator status rules, Scenario Lab invariance (30+ YAML cases), end-to-end pipeline integration, Streamlit UI smoke test, full 150-case LLM eval (blocked on API quota; 2 processed locally).

4. Plan for the remaining time

Task	Owner	Detail
Recommendation orchestrator	Kratik	Status rules RULED_OUT > REVIEW > ABSTAIN > RECOMMEND; Evidence Pack builder; 30+ YAML invariance cases.
Streamlit UI	Akshit	Customer Hub (Proof Cards, mobile-first), Cockpit (CO2e/energy/paper/digital separately; propensity × impact quadrant; Customer → Opportunity → Evidence → Action).
Scenario Lab	All	Set vehicle to EV, remove mileage, add conflict, stale data — all three verdicts recompute with exact cause displayed.
Full LLM eval (150 packs)	Madhav	Judge pipeline once API quota is available; numeric fidelity, unsupported-claim rate, fallback rate, latency per 1,000 insights.
End-to-end make all	Aniket	Clean-clone rerun, seed 42; SHA-256 artefact hash manifest; scenario recompute benchmark (target < 1 s).
Security red-team	Kratik	IDOR, role limits, SQL injection, audit-log tamper detection, secret scanning and dependency audit in CI.
Code push to main + README	All	Merge all branches to main; resolve conflicts; update README (setup, endpoints, env vars, limitations).